Lec-12

12/10/26 5:05 PM

Given an integer array `nums`, rotate the array to the right by `k` steps, where `k` is non-negative.

Example 1:

```
Input: nums = [1,2,3,4,5,6,7], k = 3
Output: [5,6,7,1,2,3,4]
Explanation:
rotate 1 steps to the right: [7,1,2,3,4,5,6]
rotate 2 steps to the right: [6,7,1,2,3,4,5]
rotate 3 steps to the right: [5,6,7,1,2,3,4]
```

1 2 3 4 5 6 7

Example 2:

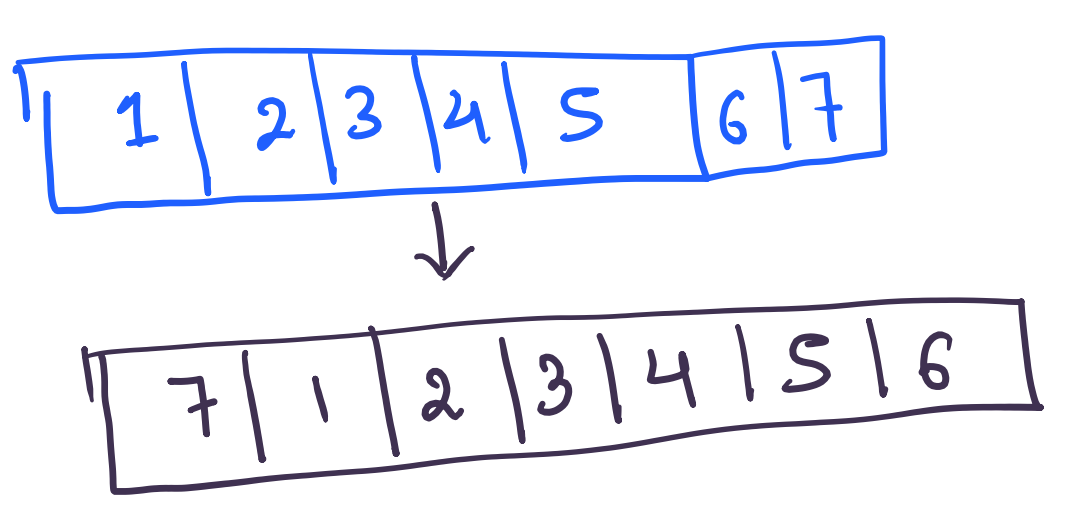
```
Input: nums = [-1,-100,3,99], k = 2
Output: [3,99,-1,-100]
Explanation:
rotate 1 steps to the right: [99,-1,-100,3]
rotate 2 steps to the right: [3,99,-1,-100]
```

[1, 2, 3, 4, 5, 6, 7]

K = 1	7 1 2 3 4 5 6
K = 2	6 7 1 2 3 4 5
K = 3	5 6 7 1 2 3 4
K = 4	4 5 6 7 1 2 3
K = 5	3 4 5 6 7 1 2
K = 6	2 3 4 5 7 6 1
K = 7	1 2 3 4 5 6 7
K = 8	7 1 2 3 4 5 6
K = 9	6 7 1 2 3 4 5
K = 10	5 6 7 1 2 3 4
K = 11	4 5 7 6 7 1 2]

① K = K%N ①

[1, 2, 3, 4, 5, 6, 7]



N = arr.len-1

① item = arr[N-1] ↳ 6

7 | 1 | 2 | 3 | 4 | 5 | 6
0 1 2 3 4 5 6

(ii) for(i = N-2 i>=0 i--){
arr[i+1] = arr[i]

i=5 arr[6] = arr[5]
i=4 arr[5] = arr[4]
i=3 arr[4] = arr[3]
i=2 arr[3] = arr[2]
i=1 arr[2] = arr[1]
i=0 arr[1] = arr[0]

(iii) arr[0] = item

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 1, 2, 3, 4, 5, 6, 7 };
	int k = 3;
	Rotate(arr, k);
	for (int i = 0; i < arr.length; i++) {
		System.out.print(arr[i] + " ");
	}

}
public static void Rotate(int[] arr, int k) {

	int n = arr.length;
	k = k % n;
	for (int j = 1; j <= k; j++) {
		int item = arr[n - 1];
		for (int i = n - 2; i >= 0; i--) {
			arr[i + 1] = arr[i];
		}
		arr[0] = item;
	}

}
```

5

K = 2%5 = 2

4 | 5 | 1 | 2 | 3
0 1 2 3 4

K = 2

item = arr[4] ↳ 4

i=3 arr[4] = arr[3]
i=2 arr[3] = arr[2]
i=1 arr[2] = arr[1]
i=0 arr[1] = arr[0]

TLE

Reversed Algo

N-K K
[1, 2, 3, 4, 5, 6, 7] → [1 2 3 4 7 6 5]
[4 3 2 1 7 6 5]

K = 3 5 6 7 1 2 3 4

5 6 7 1 2 3 4

① Starting → N-K element → i = 0, j = N-K-1
(ii) Last → K element → i = N-K j = N-1

[1, 2, 3, 4, 5, 6, 7] K = 3, N = 7

N-K = 7-3
= 4

```java
public static void Rotate(int[] arr, int k) {

	int n = arr.length;
	k = k % n;
	// 1st n-k element reverse krna hai
	Reverse(arr, 0, n - k - 1);
	// last  k element reverse krna hai
	Reverse(arr, n-k, n -  1);
	// saare element reverse krna hai
	Reverse(arr, 0, n-1);
}
```

0, 3

[1, 2, 3, 4, 5, 6, 7]
0 1 2 3 4 5 6

4 3 2 1 7 6 5

5 6 7 1 2 3 4

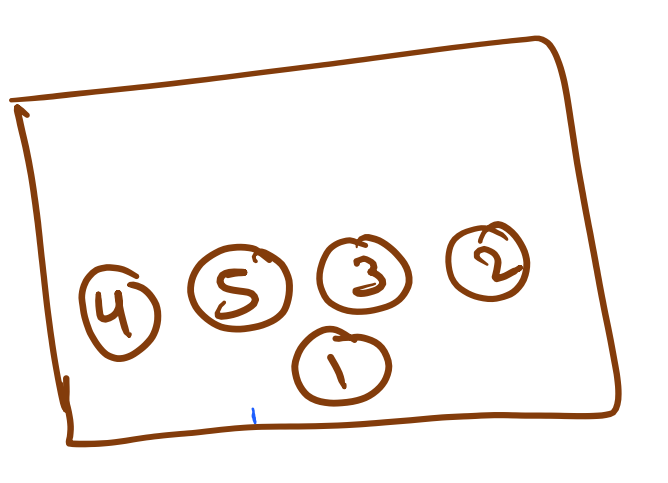
Sorting

[4 5 3 2 1] 1 2 3 4 5

- (i) bubble
- (ii) selection
- (iii) Insertion
- (iv) D.N.F
- (v) counting
- (vi) Merge sort
- (vi) Heap sort
- (vii) Quick sort
- Randomize quick sort
- cycle sort

[4, 5, 3, 2, 1]

① bubble sort [4 5 3 2 1]



① 4 5 3 2 1
4 5 3 2 1
4 3 5 2 1
4 3 2 5 1
4 3 2 1 5

(ii) 4 3 2 1 5
3 4 2 1 5
3 2 4 1 5
3 2 1 4 5

(iii) 3 2 1 4 5
2 3 1 4 5
2 1 3 4 5

(iv) 2 1 3 4 5
1 2 3 4 5

```
for(turn=1 turn<N;turn++){
 for(i=0 i<3 i++){
  if(arr[i] > arr[i+1]){
   swap(i, i+1)
  }
 }
}
```

arr.len - turn

```
for(turn=1 turn<N;turn++){
 for(i=0 i<3 i++){
  if(arr[i] > arr[i+1]){
   swap(i, i+1)
  }
 }
}
```

① [4, 5, 3, 2, 1]
0 1 2 3 4
i=0 [4 5 3 2 1]
0 1 2 3 4
i=1 [4 3 5 2 1]
0 1 2 3 4
i=2 [4, 3, 2 5 1]
0 1 2 3 4
i=3 [4 3 2 1 5]
0 1 2 3 4

(ii) [4, 3, 2, 1, 5]
i=0 [3, 4, 2, 1, 5]
0 1 2 3 4
i=1 [3, 2, 4, 1, 5]
0 1 2 3 4
i=2 [3, 2, 1, 4, 5]
0 1 2 3 4

(iii) [3, 2, 1, 4, 5]
0 1 2 3 4
i=0 [2 3 1 4 5]
0 1 2 3 4
i=1 [2, 1, 3, 4 5]

(iv) [2, 1, 3, 4, 5]
0 1 2 3 4
i=0 [1, 2 3, 4, 5]

0 1 2 3 4 5 6
[4, -1, 5, 3, 2, 1, 7]

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int[] arr = { 4, -1, 5, 3, 2, 1, 7 };
	System.out.println(min_form_ith_index(arr, 2));
}
public static int min_form_ith_index(int[] arr, int idx) {
}
```

[4, -1, 5, 3, 2, 1, 7]
0 1 2 3 ~~4~~ 4 5 6

Mini = idx → 2

```
for(i = idx+1  i<N  i++){
  if(arr[i] < arr[mini]){
    mini = i
  }
}
```